6.5 First fractions: activities

← Unit overview

1. Fraction strips

You need: six strips of paper the same length (cut from A4 lengthways), a ruler, coloured pencils, scissors.

1. Keep one strip whole. Write "1" (*één geheel*) on it.
2. Fold one strip in half. Label each part ½.
3. Fold one into quarters (half, then half again). Label ¼.
4. Fold one into eighths. Label 1/8.
5. Fold one into thirds (roll it into an S and flatten). Label 1/3.
6. Fold the last into sixths (thirds, then half). Label 1/6.

Lay them under each other. "How many quarters fit in a half? Which is bigger, 1/3 or ¼?" Keep the strips in an envelope for the rest of the unit.

Talk about it: "What happens to the size of each piece when you fold more times?"

2. Fair pizza

You need: paper circles (trace around a plate), scissors, a real pizza or pancake if you're lucky.

"Share this pizza fairly between 2 people. Now 4. Now 3." Cut the paper circles. Then the unfair version: cut a circle into 4 *unequal* pieces. "Is each piece a quarter?"

For a real pizza or pancake: let him cut it. "What fraction did you eat? What fraction is left?"

Talk about it: "Would you rather have 1/3 or ¼ of a pizza? Why?"

3. Chocolate bar fractions

You need: a chocolate bar with squares (e.g. 3 × 4 = 12 squares), or a drawing of one.

- "What fraction is one square?" (1/12)
- "Break off one row. What fraction is that?" (3 out of 12, which is also ¼ if the bar is 4 rows of 3)
- "If you eat 5 squares, what fraction is left?" (7/12)
- "Can you break off half? A third?"

Talk about it: "Can a fraction have two different names?" (3/12 and ¼.)

4. Washing-line number line

You need: a piece of string, pegs, cards.

Hang the string. Peg a card "0" at one end and "1" at the other. Give him cards with ½, ¼, ¾, 1/3, 2/3, 1/8, 1/10, 9/10 and let him peg them where they belong. Then add "2" at the far end of a longer string and include 1½.

Ask: "Which ones are close to 0? Close to 1? Close to a half?"

Talk about it: "How did you decide where 2/3 goes?"

5. Fraction of my money

You need: coins (e.g. 20 × €1 coins or 20 × 10 cent coins).

- “Here is €20. Take half. Now take a quarter.” He shares the coins into equal piles.
- “Three quarters of €20?” Three of the four piles.
- Reverse: “I have a quarter of my money here: €5. How much do I have altogether?”
- Real life: “Your pocket money is €8. If you save ¼, how much do you save?” (€2)

Talk about it: “What do you do first when you work out a fraction of an amount?”

Conversation prompts (any time)

- “Where do you see fractions at home?” (Half an hour, a quarter past, half a litre, half price.)
- “Is half of a big pizza the same as half of a small pizza?”
- “Can you show me a half in three different ways?”
- “Which is more: ½ of 10 or ¼ of 24?” (5 vs 6.)